04 MAR 2026

AADK TASK 1

Advanced Theming Strategy:
Material 3 + Brand Identity

MindMatrix

1. Research on Theming Concepts

colorScheme (Material 3)

colorScheme is a structured set of colors used throughout the app (primary, secondary, background, surface, error, etc.) to maintain consistency and accessibility.

Example:

```
private val LightColors = lightColorScheme(
    primary = Color(0xFF4A6FFF),
    secondary = Color(0xFF00BFA5),
    background = Color(0xFFF8F9FF)
)
```

typography (Material 3)

typography defines text styles like headings, body text, labels, etc., ensuring consistent font size, weight, and spacing across the app.

Example:

```
val AppTypography = Typography(
    headlineMedium = TextStyle(
        fontSize = 24.sp,
        fontWeight = FontWeight.Bold
    )
)
```

shapes (Material 3)

shapes define corner sizes and component outlines (cards, buttons, dialogs). This creates a consistent visual style.

Example:

```
val AppShapes = Shapes(
    medium = RoundedCornerShape(12.dp)
)
```

Dynamic Colour (Android 12+)

Dynamic colour automatically generates a color palette based on the user's wallpaper (Material You). It personalizes the app experience.

Example:

```
val colors = if (dynamicColor && Build.VERSION.SDK_INT >=
Build.VERSION_CODES.S) {
    dynamicLightColorScheme(context)
} else {
    LightColors
}
```

Light/Dark Theme Support

Light theme uses bright backgrounds with dark text. Dark theme uses dark backgrounds with light text. It improves readability and reduces eye strain.

Example:

```
val colors = if (isSystemInDarkTheme()) DarkColors else LightColors
```

2. Brand + Theme Design

Brand Identity

App Name: ZenTasks

Logo Idea: A minimalist checkmark inside a soft rounded square with gradient blue tones.

Brand Personality: Calm, productive, modern, clean UI.

Color Palette

Token	Hex Value
Primary	#4A6FFF
Secondary	#00BFA5
Tertiary	#FFB300
Background	#F8F9FF
Surface	#FFFFFF
Error	#B00020

Dark Theme

Token	Hex Value
Primary	#9AAEFF
Secondary	#5DF2D6
Tertiary	#FFD54F
Background	#121212
Surface	#1E1E1E
Error	#CF6679

Typography Tokens

Style	Size	Weight	Font
Heading	24sp	Bold	Sans Serif
Subtitle	18sp	Medium	Sans Serif
Body	16sp	Normal	Sans Serif
Caption	12sp	Light	Sans Serif

Example:

```
val ZenTypography = Typography(
    headlineMedium = TextStyle(
        fontSize = 24.sp,
        fontWeight = FontWeight.Bold
    ),
    bodyMedium = TextStyle(
        fontSize = 16.sp
    )
)
```

Shape Tokens

Token	Value
smallCorner	4.dp
mediumCorner	12.dp
largeCorner	24.dp

Example:

```
val ZenShapes = Shapes(
    small = RoundedCornerShape(4.dp),
    medium = RoundedCornerShape(12.dp),
    large = RoundedCornerShape(24.dp)
)
```

3. Theme Architecture Document

File Structure

```
ui/theme/
    ├── Color.kt
    ├── Typography.kt
    ├── Shapes.kt
    └── Theme.kt
```

- **Color.kt**
  Contains LightColorScheme and DarkColorScheme definitions.
- **Typography.kt**
  Contains ZenTypography definition.
- **Shapes.kt**
  Contains ZenShapes definition.
- **Theme.kt**
  Wraps everything inside MaterialTheme.

Applying Theme

```
@Composable
fun ZenTasksTheme(
    darkTheme: Boolean = isSystemInDarkTheme(),
    dynamicColor: Boolean = true,
    content: @Composable () -> Unit
) {
    val colorScheme = when {
        dynamicColor && Build.VERSION.SDK_INT >=
Build.VERSION_CODES.S ->
            if (darkTheme)
dynamicDarkColorScheme(LocalContext.current)
            else dynamicLightColorScheme(LocalContext.current)

        darkTheme -> DarkColors
        else -> LightColors
    }
    MaterialTheme(
        colorScheme = colorScheme,
        typography = ZenTypography,
        shapes = ZenShapes,
        content = content
    )
}
```

4. Theming Across UI Components

AppBar

- Background: colorScheme.primary
- Title text: colorScheme.onPrimary
- Typography: headlineMedium
- Shape: default rectangle

Mapping:
AppBar → primary color + headlineMedium typography

CardItem

- Background: colorScheme.surface
- Text: colorScheme.onSurface
- Shape: shapes.medium
- Elevation: 4.dp

Mapping:
CardItem → surface color + mediumCorner + bodyMedium text

Button

- Background: colorScheme.secondary
- Text: colorScheme.onSecondary
- Shape: shapes.large
- Typography: labelLarge

Mapping:
Button → secondary color + largeCorner + label style

5. Reflection

A well-structured theming system is very important for scalability because it keeps the UI consistent across all screens. If colors and fonts are hard-coded in multiple places, changing the brand design later becomes very difficult and time-consuming. It can also cause visual inconsistency and poor user experience.

Using dynamic colour and light/dark themes improves personalization and accessibility. Users feel the app matches their device style, and dark mode reduces eye strain at night. Overall, proper theming makes the app more professional, maintainable, and future-proof.